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# exdqlm: An R Package for Estimation and Analysis of Flexible Dynamic Quantile Linear Models

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**Abstract**

We present the R package **exdqlm** for Bayesian quantile regression, with primary emphasis on dynamic state-space quantile models for time series. The package is built around flexible dynamic quantile linear models (exDQLMs), which use the extended asymmetric Laplace (exAL) family, a parametric extension of the asymmetric Laplace (AL) commonly used in quantile regression. The software provides exact posterior inference via Markov chain Monte Carlo (MCMC) and fast approximate posterior inference via variational Bayes (VB), supporting posterior uncertainty quantification while also providing a computationally efficient option for longer time series. It also supports static exAL quantile regression with regularized priors, transfer-function models for nonlinear input effects at a given quantile, post hoc posterior-predictive synthesis across separately fitted quantiles, forecasting, and several quantitative and visual diagnostics for model evaluation.

*Keywords*: dynamic quantile regression, dynamic models, asymmetric Laplace, variational Bayes, coordinate ascent variational inference, quantile regression.

## 1. Introduction

Quantile regression has become a widely used alternative to mean-based modeling in static settings, but dynamic quantile methods and accompanying software remain comparatively limited. In Bayesian parametric quantile regression, the asymmetric Laplace (AL) likelihood has played a central role because it is linked to the check-loss function used in classical quantile regression (Yu and Moyeed 2001; Koenker 2005) and admits convenient mixture representations for posterior computation. These computational advantages have made AL-

based models attractive, but the AL can be restrictive as an error model. More flexible error distributions have been studied extensively in the Bayesian nonparametric literature (Kottas and Krnjajić 2009; Reich, Bondell, and Wang 2009; Taddy and Kottas 2010); however, the resulting methods can be computationally demanding for dynamic models and longer time series. Yan, Zheng, and Kottas (2025) introduce the extended asymmetric Laplace (exAL), a parametric extension of the AL that preserves its computational advantages while relaxing some of its restrictive features. Barata, Prado, and Sansó (2022) use this family to develop a flexible dynamic quantile linear model (exDQLM), which is the model implemented in the R package **exdqlm**.

The package **exdqlm** provides a unified workflow for Bayesian inference, forecasting, and model assessment for quantile models fitted one quantile at a time. Its core functionality is dynamic state-space quantile modeling, supported by exact posterior simulation via MCMC and a faster variational Bayes (VB) approximation. The package also includes transfer-function exDQLMs for nonlinear input effects at a given quantile, static exAL quantile regression with shrinkage priors, and a synthesis step that combines anchor-specific posterior predictive draws from separately fitted quantile models into a unified posterior predictive distribution. Forecasting and a range of visual and quantitative diagnostics are available throughout. The package is released under the MIT License and is available from the Comprehensive R Archive Network (CRAN) at `https://CRAN.R-project.org/package=exdqlm`.

The current R software ecosystem for quantile regression is broad, but the capabilities most relevant to this paper are distributed across software with different aims. The frequentist package **quantreg** (Koenker 2025) remains the broad general-purpose baseline, covering linear, nonlinear, nonparametric, censored, and dynamic quantile regression, together with bootstrap-based and related uncertainty summaries. Its `dynrq()` function supports dynamic linear quantile regression through time-series regression formulas involving lags, differences, trends, seasonal factors, and harmonic terms, while preserving time-series attributes. However, the dynamic structure enters through the design matrix rather than through latent states: once these regressors are constructed, estimation proceeds through the standard **quantreg** fitting routines for a fixed coefficient vector. In that sense, `dynrq()` is a regression-style time-series interface rather than a latent-state state-space model, so it does not provide filtered or smoothed state summaries.

Most other R packages relevant here are static, or else they target a different multi-quantile objective. Among the static Bayesian options, **bayesQR** (Benoit, Al-Hamzawi, Yu, and Van den Poel 2023) provides Bayesian quantile regression under the AL working likelihood with an adaptive-lasso option, whereas **pqrBayes** (Fan, Wu, Ren, Li, and Zhou 2026) emphasizes penalized Bayesian quantile regression with spike-and-slab and horseshoe-family shrinkage priors. The broader Bayesian modeling package **brms** (Bürkner 2026) is also relevant because it supports AL-based quantile regression within a flexible multilevel, nonlinear, and additive regression framework, and can incorporate some autocorrelation structures. However, it is not organized as a dedicated quantile state-space workflow, and its documentation does not present filtered or smoothed state summaries of the type targeted by **exdqlm**. Packages such as **qrjoint** (Tokdar and Cunningham 2025), **quantregGrowth** (Muggeo 2025), and **qrcm** (Frumento 2025) instead address a different cross-quantile objective through joint estimation, non-crossing constraints, or coefficient modeling across quantiles. Other packages focus on specialized static settings, including additive or semiparametric modeling in **qgam** (Fasiolo and Griffiths 2025), penalized frequentist fitting across quantiles in **rqPen** (Sherwood, Li, and

| Package | Primary focus | Temporal structure | Fast approximate inference | Cross-quantile treatment |
|---|---|---|---|---|
| **quantreg** | Broad frequentist quantile regression | Regression-style dynamic via `dynrq()` | No | Post hoc rearrangement |
| **bayesQR** | Static Bayesian quantile regression under the AL working likelihood | Static | No | None |
| **pqrBayes** | Static Bayesian penalized quantile regression | Static | No | None |
| **qrjoint** | Joint estimation of non-crossing quantiles in regression settings | Static | No | Joint non-crossing estimation |
| **qgam** | Additive and semiparametric quantile regression | Static | No | No dedicated joint/non-crossing treatment |
| **exdqlm** | Bayesian dynamic/state-space quantile regression; static exAL regression | Latent-state state-space | Yes | Post hoc predictive synthesis |

Table 1: Representative R packages spanning broad, static Bayesian, additive, joint multi-quantile, and dynamic/state-space quantile-regression workflows. Temporal structure distinguishes static packages, regression-style dynamic formulations, and latent-state state-space models. Fast approximate inference denotes a package-facing deterministic posterior approximation, such as variational Bayes. Cross-quantile treatment records any package-specific way of linking information across quantile levels; these entries are not methodologically interchangeable. In particular, post hoc predictive synthesis is distinct from joint non-crossing estimation.

Maidman 2026), convolution-smoothed linear quantile regression with efficient gradient-based fitting in **conquer** (He, Pan, Tan, and Zhou 2023), and robust or censored-response modeling in **lqr** (Galarza, Benites, Bourguignon, and Lachos 2024).

Table 1 groups a small set of packages by the software distinctions most relevant for situating **exdqlm** within the current quantile-regression ecosystem.

Related implementations also exist outside R. In Python, **statsmodels** (statsmodels developers 2025) provides linear quantile regression via iterative reweighted least squares. Julia offers **QuantReg.jl** (Fogarty 2020), and MATLAB includes tools such as `fitrqlinear` and `fitrqnet` (MathWorks 2026). These implementations are useful reference points for the broader software ecosystem, but they are better viewed as general or static quantile-regression tools than as direct counterparts to a dedicated package for Bayesian dynamic quantile state-space modeling.

Viewed against this software ecosystem, **exdqlm** is not intended to replace broad general-purpose quantile-regression toolkits, additive quantile-regression packages, or software designed specifically for joint multi-quantile estimation. Rather, it addresses a more specific need: a unified Bayesian workflow for dynamic/state-space quantile modeling that combines exact posterior simulation with a fast variational approximation, while also using the more flexible exAL likelihood, supporting static exAL regression with shrinkage priors ranging from ridge to regularized horseshoe variants, and providing transfer-function models, forecasting, diagnostics, and post hoc posterior-predictive synthesis from separately fitted quantile mod-

els. General state-space packages such as **dlm** (Petris and Gilks 2018) and **dynr** (Ou, Hunter, Chow, Ji, Chen, Hung, Lee, Li, and Park 2021) remain useful for constructing dynamic components, but they do not provide dedicated quantile-regression workflows.

The remainder of the paper is organized as follows. In Section 2, we describe the modeling framework, including the algorithms used for posterior inference and our methods for including nonlinear relationships. We also discuss static quantile regression as a special case of the framework. In Section 3 we detail the model diagnostics and forecasting method implemented in the package. Then, in Section 4, we walk through four step-by-step examples on how to use the package. Finally, in Section 5, we present a summary of the package capabilities.

## 2. Extended dynamic quantile linear models

Consider a set of scalar time-evolving responses, $y_t$, for times $t = 1, \ldots, T$. For each $t$, an extended dynamic quantile linear model (exDQLM) for inference on a single $p_0$-th quantile is defined by

$$\text{Observation equation:} \quad y_t = \mathbf{F}_t'\boldsymbol{\theta}_t + \epsilon_t, \qquad \epsilon_t \sim \text{exAL}_{p_0}(0, \sigma, \gamma) \tag{1}$$

$$\text{System equation:} \quad \boldsymbol{\theta}_t = \mathbf{G}_t\boldsymbol{\theta}_{t-1} + \boldsymbol{\omega}_t, \qquad \boldsymbol{\omega}_t \sim \text{N}(\mathbf{0}, \mathbf{W}_t). \tag{2}$$

Here $\mathbf{F}_t$ is the $q \times 1$ regression vector of the covariates corresponding to the $q \times 1$ parameter vector $\boldsymbol{\theta}_t$ at time $t$, $\mathbf{G}_t$ is the $q \times q$-dimensional evolution matrix defining the structure of the parameter vector evolution in time, and the normal distribution according to which the system error $\boldsymbol{\omega}_t$ evolves is $q$-variate. The observational errors $\epsilon_t$ of the exDQLM follow an exAL distribution for fixed quantile $p_0$, whose density is denoted as $\text{exAL}_{p_0}(\cdot)$, and it is such that $\int_{-\infty}^{0} \text{exAL}_{p_0}(\epsilon_t|0, \sigma, \gamma)\text{d}\epsilon_t = p_0$. Thus, the observation equation in Equation (1) implies that $\mathbf{F}_t'\boldsymbol{\theta}_t$ corresponds to the $p_0$-quantile of $y_t$, even if this is not made explicit in the notation above. Table 2 shows the function parameters that define the structure of the dynamic quantile, specifically $\mathbf{F}_t$ and $\mathbf{G}_t$ for $t = 1, \ldots, T$.

The exAL distribution, introduced by Yan *et al.* (2025), extends the location-scale mixture representation of the AL (Kozumi and Kobayashi 2011). For fixed $p_0$, the skewness parameter $\gamma$ is restricted to an interval $(L, U)$, where $L$ and $U$ are the negative and positive roots of $g(\gamma) = 1 - p_0$ and $g(\gamma) = p_0$, respectively, with $g(\gamma) = 2\Phi(-|\gamma|)\exp(\gamma^2/2)$. The package also exports direct distributional utilities for the exAL family. These include `dexal()`, `pexal()`, `qexal()`, and `rexal()`, together with `get_gamma_bounds()` for recovering this admissible interval. This results in a mixture representation of $\text{exAL}_{p_0}(0, \sigma, \gamma)$ as follows:

$$\int\int_{\mathbb{R}^+\times\mathbb{R}^+} \text{N}(y_t|C(p,\gamma)\sigma|\gamma|s_t + A(p)v_t, \sigma B(p)v_t)\text{Exp}(v_t|\sigma)\text{N}^+(s_t|0,1)\text{d}v_t\text{d}s_t. \tag{3}$$

Here, $p = p(p_0, \gamma) = I(\gamma < 0) + \{[p_0 - I(\gamma < 0)]/g(\gamma)\}$ where $g(\gamma) = 2\Phi(-|\gamma|)\exp(\gamma^2/2)$, and $\Phi(\cdot)$ denotes the standard normal CDF. $A(p)$, $B(p)$, $C(p, \gamma)$ are the functions of $p$ and $\gamma$: $A(p) = \frac{1-2p}{p(1-p)}$, $B(p) = \frac{2}{p(1-p)}$, $C(p, \gamma) = [I(\gamma > 0) - p]^{-1}$. Lastly, $\text{Exp}(v|\sigma)$ denotes the exponential distribution with mean $\sigma$, and $\text{N}^+(s_t|0, 1)$ denotes a normal distribution truncated to the positive reals with mean 0 and variance 1. This mixture representation of the exAL

| Input list | `model` | | | | `PriorSigma` | | `PriorGamma` | | |
|---|---|---|---|---|---|---|---|---|---|
| Symbol | $\mathbf{F}_{1:T}$ | $\mathbf{G}_{1:T}$ | $\mathbf{m}_0$ | $\mathbf{C}_0$ | $a_\sigma$ | $b_\sigma$ | $m_\gamma$ | $s_\gamma$ | $\nu_\gamma$ |
| List component | `FF` | `GG` | `m0` | `C0` | `a_sig` | `b_sig` | `m_gam` | `s_gam` | `df_gam` |
| Default | - | - | $\mathbf{0}_q$ | 1e3$\mathbf{I}_q$ | 2.1 | 1.1 | 0 | 1 | 1 |

Table 2: Translation from mathematical symbols to parameters used by **exdqlm** functions, as well as default values where applicable. $\mathbf{0}_q$ denotes a $q$-dimensional vector of zeros, and $\mathbf{I}_q$ denotes the $q$-dimensional identity matrix.

can be exploited to rewrite the exDQLM as the following hierarchical model for $t = 1, \ldots, T$:

$$y_t|\boldsymbol{\theta}_t, \sigma, \gamma, v_t, s_t \sim \mathrm{N}(y_t|\mathbf{F}_t'\boldsymbol{\theta}_t + C(p,\gamma)\sigma|\gamma|s_t + A(p)v_t, \sigma B(p)v_t) \tag{4}$$

$$v_t, s_t|\sigma \sim \mathrm{Exp}(v_t|\sigma)N^+(s_t|0,1) \tag{5}$$

$$\boldsymbol{\theta}_t|\boldsymbol{\theta}_{t-1}, \mathbf{W}_t \sim \mathrm{N}(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}_t). \tag{6}$$

This hierarchical form is leveraged for the posterior inference implemented within the **exdqlm** package.

## 2.1. Prior specification

Under a Bayesian formulation and given $p_0$, we specify priors for the initial state vector $\boldsymbol{\theta}_0$, scale parameter $\sigma$ and skewness parameter $\gamma$. For $\boldsymbol{\theta}_0$, we assume a $q$-variate conjugate normal prior $\boldsymbol{\theta}_0 \sim \mathrm{N}(\mathbf{m}_0, \mathbf{C}_0)$. For $\sigma$, we assume a conjugate inverse-gamma prior denoted as $\mathrm{IG}(a_\sigma, b_\sigma)$. In many cases a strong prior on $\sigma$ is necessary to guarantee fast posterior convergence and estimation. This is illustrated in the examples of Section 4. The parameter $\gamma$ has bounded support over the interval $(L, U)$ where $L$ is the negative root of $g(\gamma) = 1 - p_0$ and $U$ is the positive root of $g(\gamma) = p_0$ (Yan *et al.* 2025). In contrast to the flat prior suggested by Yan *et al.* (2025), we find that using a proper prior on the skewness parameter promotes reliable posterior inference. Thus, we place a Student-t distribution truncated to the interval $(L, U)$ as the prior for $\gamma$, i.e. $\gamma \sim \mathrm{t}_{(L,U)}(m_\gamma, s_\gamma)$ with $\nu_\gamma$ degrees of freedom. The prior parameters used in the **exdqlm** functions, their corresponding mathematical symbols, and default values are shown in Table 2.

## 2.2. Posterior estimation

The hierarchical representation of the exDQLM facilitates posterior simulation using Markov Chain Monte Carlo (MCMC) algorithms. This is implemented in the function `exdqlmMCMC()`. Gibbs sampling steps are used to update the sets of latent variables $s_t$ and $v_t$ for $t = 1, \ldots, T$. The dynamic regression coefficients are sampled using a forward filtering backwards sampling algorithm (Carter and Kohn 1994; Frühwirth-Schnatter 1994). The scale and skewness parameters are updated separately from the latent-state blocks. By default, the package uses a slice-sampling update for the $(\sigma, \gamma)$ block, while joint random-walk Metropolis-Hastings (MH) alternatives are also available. When desired, the chain can be warm-started from the variational approximation; this initialization is separate from the subsequent MCMC proposal kernel. An example of this functionality can be found in Section 4.1. We perform a total of $N_{BURN} + N_{MCMC}$ MCMC iterations, discarding the first $N_{BURN}$ and retaining the remaining $N_{MCMC}$ posterior samples. The object created by `exdqlmMCMC()` includes posterior samples

| **exdqlm** function | `exdqlmMCMC()` | | | | `exdqlmLDVB()` | |
|---|---|---|---|---|---|---|
| Symbol | $\mathcal{K}_{\sigma,\gamma}$ | $\Sigma_{MH}$ | $N_{BURN}$ | $N_{MCMC}$ | $\epsilon_{tol}$ | $N_{SAMP}$ |
| Input | `mh.proposal` | `Sig.mh` | `n.burn` | `n.mcmc` | `tol` | `n.samp` |
| Default | `"slice"` | $0.05\mathbf{I}_2$ | 2000 | 1500 | 0.1 | 200 |

Table 3: Translation from mathematical symbols to parameters used in functions `exdqlmMCMC()` and `exdqlmLDVB()`, as well as default values.

of all parameters $(\boldsymbol{\theta}_{1:T}, v_{1:T}, s_{1:T}, \gamma, \sigma)$, filtered and smoothed summaries for the state vector $\boldsymbol{\theta}_{1:T}$, and diagnostics for the proposal kernel used in the $(\sigma, \gamma)$ update.

For purposes of model selection or large time series data sets, posterior sampling with the MCMC algorithm can be computationally demanding. To address this, the package includes a variational Bayes (VB) routine for fast approximate posterior estimation. The default routine, `exdqlmLDVB()`, uses a mean-field approximation together with a Laplace-Delta treatment of the non-conjugate $(\sigma, \gamma)$ block. The variational Bayes (VB) framework approximates the posterior distribution with partitioned variational distributions (Ostwald, Kirilina, Starke, and Blankenburg 2014; Beal 2003). Under a mean-field factorization, this induces a blockwise coordinate-ascent variational inference scheme. For the exDQLM, the blocks associated with the dynamic coefficients and latent variables admit recognizable closed-form updates, while the joint variational distribution of $\sigma$ and $\gamma$ is non-conjugate. Following the non-conjugate VB strategy of Wang and Blei (2013), we approximate this block on transformed coordinates and use a second-order Delta method to evaluate the expectations of functions of $(\sigma, \gamma)$ needed by the remaining variational updates. The dynamic regression coefficients are updated within each iteration using a forward filtering backwards smoothing algorithm. Convergence is assessed jointly through the state, scale, skewness, and ELBO diagnostics. After the algorithm reaches convergence with tolerance $\epsilon_{tol}$, $N_{SAMP}$ samples are drawn from the variational distributions, resulting in an approximate sample of the posterior distribution. The output of `exdqlmLDVB()` includes the approximate posterior samples as well as the parameters of the variational distributions for all parameters $(\boldsymbol{\theta}_{1:T}, v_{1:T}, s_{1:T}, \gamma, \sigma)$. For the $(\gamma, \sigma)$ block, the return also contains the transformed mode, approximate covariance matrix, derived moments, and ELBO terms associated with the Laplace-Delta approximation. Selected state-space computations in the dynamic routines are implemented in C++, with corresponding R code retained for parity checks and fallback use.

Table 3 shows the input parameters of the fitting functions `exdqlmMCMC()` and `exdqlmLDVB()`, their mathematical symbols, and their default values.

The argument `Sig.mh` is only used when `mh.proposal` selects a joint random-walk MH kernel; it is not used under the default slice-sampling update.

### 2.3. Specification of the evolution covariance via discount factors

Both estimation algorithms, `exdqlmMCMC()` and `exdqlmLDVB()`, utilize discount factors to specify the time-evolving covariance matrices $\mathbf{W}_t$ introduced in Equation (2) (West, Harrison, and Migon 1985). More precisely, we define the evolution variance matrix $\mathbf{W}_t = \frac{1-\delta}{\delta}\mathbf{G}_t\mathbf{C}_{t-1}\mathbf{G}_t'$ where $\mathbf{C}_{t-1}$ denotes the posterior variance of the state vector $\boldsymbol{\theta}_{t-1}$ at time $t-1$ and $\delta$ is a discount factor such that $0 < \delta \leq 1$. The discount factor $\delta$ can be specified

by the user via function parameter `df`. Selection of discount factors is typically done by optimizing a model-checking criterion. In practice we recommend predictive criteria such as the CRPS or pplc discussed in Section 3, with the KL divergence retained as a complementary calibration diagnostic. A brief example of this method for discount factor selection can be found in Section 4.2.

Discounting by component of the state-space model is also a functionality of our estimation algorithms. Suppose the state vector is comprised of $h$ components $\boldsymbol{\theta}_{it}$ of dimension $q_i$ for $i = 1, \ldots, h$ such that $\boldsymbol{\theta}'_t = (\boldsymbol{\theta}'_{1t}, \ldots, \boldsymbol{\theta}_{ht})'$ and $\sum_{i=1}^{h} q_i = q$. If $\mathbf{W}_{it}$ denotes the $i^{th}$ component of the evolution covariance matrix such that $\mathbf{W}_t = \text{blockdiag}\{\mathbf{W}_{1t}, \ldots, \mathbf{W}_{ht}\}$ and $\delta_i$ denotes a discount factor for the $i^{th}$ component, we can define the evolution variance matrix by component as $\mathbf{W}_{it} = \frac{1-\delta_i}{\delta_i}\mathbf{G}_{it}\mathbf{C}_{i,t-1}\mathbf{G}'_{it}$. Here $\mathbf{G}_{it}$ and $\mathbf{C}_{i,t-1}$ denote the $i^{th}$ components of $\mathbf{G}_t$ and $\mathbf{C}_{t-1}$, respectively. The function parameters `df` and `dim.df` allow users to specify a set of component discount factors $(\delta_1, \ldots, \delta_h)$ and their dimensions $(q_1, \ldots, q_h)$, respectively, within the state-space model. Examples of this feature can be found in Section 4. For a full review of discount factors including discounting by component, see West *et al.* (1985).

### 2.4. Transfer function model

Quantifying the relationship between an input and a quantile of a response is a non-trivial task, particularly when the relationship is nonlinear. In mean-centric dynamic modeling, transfer functions are a straightforward method to incorporate variables which measure the combined effect of current and past regression effects (West *et al.* 1985). Within **exdqlm**, this extension is handled by augmenting the dynamic state-space representation conditional on a fixed transfer-function rate $\lambda$, and the resulting workflow is available through both VB and MCMC fitting routines.

For $t = 1, \ldots, T$ and transfer input vector $\mathbf{x}_t \in \mathbb{R}^k$, a transfer-function exDQLM with exponential decay is defined by

$$y_t|\boldsymbol{\theta}_t, \zeta_t, \gamma, \sigma \sim \text{exAL}_{p_0}(\mathbf{F}'_t\boldsymbol{\theta}_t + \zeta_t, \sigma, \gamma) \tag{7}$$

$$\boldsymbol{\theta}_t|\boldsymbol{\theta}_{t-1}, \mathbf{W}^{\theta}_t \sim \text{N}(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}^{\theta}_t) \tag{8}$$

$$\zeta_t|\zeta_{t-1}, \boldsymbol{\psi}_{t-1}, \omega_t \sim \text{N}(\lambda\zeta_{t-1} + \mathbf{x}'_t\boldsymbol{\psi}_{t-1}, \omega_t) \tag{9}$$

$$\boldsymbol{\psi}_t|\boldsymbol{\psi}_{t-1}, \mathbf{W}^{\psi}_t \sim \text{N}(\boldsymbol{\psi}_{t-1}, \mathbf{W}^{\psi}_t). \tag{10}$$

Here $\zeta_t \in \mathbb{R}$ denotes the accumulated transfer effect on the quantile at time $t$, while $\boldsymbol{\psi}_t \in \mathbb{R}^k$ is the vector of instantaneous transfer coefficients associated with the components of $\mathbf{x}_t$. The mean contribution of the transfer block is therefore $\lambda\zeta_{t-1} + \mathbf{x}'_t\boldsymbol{\psi}_{t-1}$: the inner product $\mathbf{x}'_t\boldsymbol{\psi}_{t-1}$ captures the contemporaneous effect of the current inputs, and the parameter $\lambda \in [0, 1]$ controls how much of the previous accumulated effect is propagated forward. For $0 < \lambda < 1$, this propagated contribution decays geometrically. More explicitly, the contribution of $\mathbf{x}_t$ to the quantile at time $t + h$ is $\lambda^h\mathbf{x}'_t\boldsymbol{\psi}_{t-1}$. Therefore, for each $t$ and tolerance $\epsilon > 0$, any

$$k_t \geq \frac{\log(\epsilon) - \log(|\mathbf{x}'_t\boldsymbol{\psi}_{t-1}|)}{\log(\lambda)}$$

provides a lower bound on the number of time steps until the magnitude of the propagated effect falls below $\epsilon$. The package reports the median of this bound over time for $\epsilon = 1\text{e}{-3}$.

| Symbol | $\mathbf{x}_t$ | $\tilde{\lambda}$ | $(\delta_\zeta, \boldsymbol{\delta}_\psi)$ | $\mathbf{m}_0^{tf}$ | $\mathbf{C}_0^{tf}$ |
|---|---|---|---|---|---|
| transfer-function inputs | `X` | `lam` | `tf.df` | `tf.m0` | `tf.C0` |

Table 4: Translation from mathematical symbols to parameters used in the transfer-function fitting routines.

Conditional on a fixed $\lambda$, say $\tilde{\lambda}$, we augment the dynamic fitting routines to incorporate the transfer-function structure by replacing $\mathbf{F}_t$, $\boldsymbol{\theta}_t$, $\mathbf{G}_t$, and $\mathbf{W}_t^\theta$ in Equations (1)-(2) with

$$\tilde{\mathbf{F}}_t' = (\mathbf{F}_t', 1, \mathbf{0}_k'), \qquad \tilde{\boldsymbol{\theta}}_t' = (\boldsymbol{\theta}_t', \zeta_t, \boldsymbol{\psi}_t'),$$

$$\tilde{\mathbf{G}}_t = \text{blockdiag}\{\mathbf{G}_t, \mathbf{G}_t^{tf}\}, \qquad \mathbf{G}_t^{tf} = \begin{pmatrix} \tilde{\lambda} & \mathbf{x}_t' \\ \mathbf{0}_k & \mathbf{I}_k \end{pmatrix},$$

and

$$\tilde{\mathbf{W}}_t = \text{blockdiag}\{\mathbf{W}_t^\theta, \omega_t, \mathbf{W}_t^\psi\}.$$

This augmentation is implemented in the functions `exdqlmTransferLDVB()` and `exdqlmTransferMCMC()`, which extend the routines found in `exdqlmLDVB()` and `exdqlmMCMC()`.

Similar to the specification of $\mathbf{W}_t^\theta$ discussed in Section 2.3, discount factors are used to specify the transfer block variances $\omega_t$ and $\mathbf{W}_t^\psi$. In the package, `tf.df` can be supplied as a single shared discount factor, as a pair $(\delta_\zeta, \delta_\psi)$ with a shared value for the whole $\boldsymbol{\psi}_t$ block, or componentwise across $(\zeta_t, \psi_{1,t}, \ldots, \psi_{k,t})$. A conjugate normal prior on $(\zeta_0, \boldsymbol{\psi}_0')' \sim N(\mathbf{m}_0^{tf}, \mathbf{C}_0^{tf})$, with $\mathbf{m}_0^{tf} \in \mathbb{R}^{k+1}$ and $\mathbf{C}_0^{tf} \in \mathbb{R}^{(k+1)\times(k+1)}$, completes the transfer-function model extension. Table 4 shows the additional parameters used in the transfer-function fitting routines, as well as their mathematical symbols discussed in this section.

## 2.5. Special cases, including static quantile regression

The first special case of our exDQLM arises from the fact that the exAL is an extension of the AL (Yan *et al.* 2025). When $\gamma = 0$, the observational error $\epsilon_t$ in Equation (1) reduces such that $\epsilon_t \sim \text{exAL}_{p_0}(\epsilon_t|0, \sigma, \gamma) = \text{AL}_{p_0}(\epsilon_t|0, \sigma)$. This special case with observational errors distributed independently from an AL is the dynamic quantile linear model (DQLM) presented in Gonçalves, Migon, and Bastos (2017). The DQLM can be implemented with our package using the parameter `dqlm.ind = TRUE`.

Non-time-varying quantile regression is also a special case of our methodology. More specifically, when discount factors are set to 1, the evolution covariance $\mathbf{W}_t$ reduces to $\mathbf{0}$ resulting in non-time-varying parameters (West *et al.* 1985). Thus, the model reduces to the static exAL regression methodology of Yan *et al.* (2025). The static exAL special case is implemented directly in the package. This model can be implemented using our functions `exalStaticMCMC()` or `exalStaticLDVB()`, which provide MCMC and VB algorithms, respectively.

The final special case is the intersection of the previous two. A static quantile regression model with observational errors distributed according to an AL corresponds to Bayesian linear quantile regression under the AL working likelihood, first presented in Yu and Moyeed (2001). This methodology, utilized to create the Bayesian package for quantile regression **bayesQR**, can also be implemented with our package using functions `exalStaticMCMC()` or

`exalStaticLDVB()` with parameter `dqlm.ind = TRUE`, which acts as a convenience flag fixing $\gamma = 0$.

Examples of these special cases can be found in Section 4. A static quantile regression example specifically can be found in Section 4.4. For further details on the methodology outlined in this section, see Barata *et al.* (2022).

## 3. Model Diagnostics and Forecasting

To evaluate the quantile inference and predictive performance of the exDQLM, several quantitative and visual model diagnostics are included in the package.

The one-step-ahead predictive distribution function introduced by Rosenblatt (1952) is a useful diagnostic for models in which the marginal forecast distributions are unrecognizable or difficult to compute (Huerta, Jiang, and Tanner 2003; Prado, Molina, and Huerta 2006). For the exDQLM, this distribution is defined as

$$u_t = \Pr(Y_t \leq y_t | y_{1:t-1}, v_{1:T}, s_{1:T}, \sigma, \gamma), \tag{11}$$

with $y_{1:t-1} = y_1, \ldots, y_{t-1}$, $v_{1:T} = (v_1, \ldots, v_T)$ and $s_{1:T} = (s_1, \ldots, s_T)$. Here $u_t$ defines an independent sequence which is uniformly distributed on the interval $(0, 1)$ (Rosenblatt 1952). Conditional on $\{v_{1:T}, s_{1:T}, \sigma, \gamma\}$, the forecast distribution of $y_t$ within the exDQLM is normally distributed. Therefore, conditional on MAP estimates $\{\hat{v}_{1:T}, \hat{s}_{1:T}, \hat{\sigma}, \hat{\gamma}\}$ we estimate the sequence as

$$\hat{u}_t = \Phi(y_t | y_{1:t-1}, \hat{v}_{1:T}, \hat{s}_{1:T}, \hat{\sigma}, \hat{\gamma}) \tag{12}$$

where $\Phi$ denotes the normal CDF.

Using this estimated sequence $\{\hat{u}_t\}$, we can assess the model performance in several ways. First, we visually examine the correlation of the estimated sequence via an ACF plot. Second, by transforming the values with a standard normal inverse CDF, we compare their distributional shape to that of a standard normal with a normal QQ-plot. To quantify the divergence of this transformed sequence from normality we compute the KL divergence (Kullback and Leibler 1951), $\mathrm{KL}(h, \phi) = \int_{-\infty}^{\infty} h(x) \log \frac{h(x)}{\phi(x)} \mathrm{d}x$, where $h$ denotes the numerically approximated density of our transformed sample and $\phi$ is the standard normal density. A smaller KL divergence suggests a superior model fit. Lastly, the MAP standardized forecast errors,

$$\frac{y_t - \mathsf{E}[y_t | y_{1:t-1}, \hat{v}_{1:T}, \hat{s}_{1:T}, \hat{\sigma}, \hat{\gamma}]}{\sqrt{\mathrm{Var}(y_t | y_{1:t-1}, \hat{v}_{1:T}, \hat{s}_{1:T}, \hat{\sigma}, \hat{\gamma})}} \tag{13}$$

can also be used as an additional graphical aid in examining the distribution at specific time points. These MAP standardized forecast errors are included in the output of the `exdqlmMCMC()` and `exdqlmLDVB()` functions.

To connect the predictive criteria used below, let $\rho_p(x) = x[p - I(x < 0)]$ denote the check loss at level $p$. In addition to the calibration-based diagnostics above, we compute the mean continuous ranked probability score (CRPS) from the posterior predictive draws (Gneiting and Raftery 2007). For a predictive distribution $G$ with quantile function $G^{-1}$ and an observation $y$, the CRPS is

$$\mathrm{CRPS}(y, G) = \mathbb{E}_{Y \sim G} |Y - y| - \frac{1}{2} \mathbb{E}_{Y, Y' \sim G} |Y - Y'| = 2 \int_0^1 \rho_p\Big(y - G^{-1}(p)\Big) \, dp,$$

| Symbol | $\tilde{t}$ | $k$ | $\mathbf{F}_{(\tilde{t}+1):(\tilde{t}+k)}$ | $\mathbf{G}_{(\tilde{t}+1):(\tilde{t}+k)}$ |
|---|---|---|---|---|
| `exdqlmForecast()` input | `start.t` | `k` | `fFF` | `fGG` |

Table 5: Translation from mathematical symbols to additional parameters used in function `exdqlmForecast()`.

This representation shows that the CRPS aggregates check-loss performance over all quantile levels, so it evaluates the full predictive distribution rather than a single target quantile. It rewards forecasts that are both sharp and well calibrated, and it reduces to the absolute error, and hence to the MAE after averaging, for deterministic forecasts. Smaller values indicate better predictive performance.

For a final diagnostic, we include Gelfand and Ghosh (1998) posterior predictive loss criterion (pplc), which evaluates the posterior predictive distribution under the target-level check loss $\rho_{p_0}$. That is,

$$\text{pplc} = \sum_t \mathsf{E}[\rho_{p_0}(y_t^{obs} - y_t^{rep})|y_{1:T}], \tag{14}$$

where the expectation is with respect to the posterior predictive distribution of $y_t$, $p(y_t^{rep}|y_{1:T})$. This distribution is used to generate a replicate dataset and is the expected value of the observation equation (1) with respect to the posterior distributions of the parameters. Thus, the CRPS summarizes the full predictive distribution through an integrated check loss, whereas the pplc keeps the target quantile explicit through $\rho_{p_0}$. Smaller pplc values suggest a superior model fit. Samples from the posterior replicate distributions are included in the output of the `exdqlmMCMC()` and `exdqlmLDVB()` functions.

The function `exdqlmDiagnostics()` implements the model checking criteria discussed in this section. When an output from either `exdqlmMCMC()` and `exdqlmLDVB()` is passed to `exdqlmDiagnostics()`, the resulting object includes: the estimated sequence $\{\hat{u}_t\}$, the KL divergence, the mean CRPS, pplc, the ordered pairs of the QQ-plot, and autocorrelations by lag. By default, the function also produces the QQ-plot, ACF plot and MAP standard forecast error plot. Examples of the utility of `exdqlmDiagnostics()` can be found in the following Section 4.

### 3.1. k-step-ahead forecast

For an arbitrary time $\tilde{t}$, the $k$-step-ahead future marginal distribution of the quantile is

$$\mathbf{F}'_{\tilde{t}+k}\boldsymbol{\theta}_{\tilde{t}+k}|y_1,\ldots,y_{\tilde{t}} \sim \mathrm{N}(\mathbf{F}'_{\tilde{t}+k}a_{\tilde{t}}(k), \mathbf{F}'_{\tilde{t}+k}\mathbf{R}_{\tilde{t}}(k)\mathbf{F}_{\tilde{t}+k}) \tag{15}$$

where $\mathbf{a}_{\tilde{t}}(k) = \mathbf{G}_{\tilde{t}+k}\mathbf{a}_{\tilde{t}}(k-1)$, $\mathbf{R}_{\tilde{t}}(k) = \mathbf{G}_{\tilde{t}+k}\mathbf{R}_{\tilde{t}}(k-1)\mathbf{G}'_{\tilde{t}+k} + \mathbf{W}_{\tilde{t}+k}$, $\mathbf{a}_{\tilde{t}}(0) = \mathbf{m}_{\tilde{t}}$, and $\mathbf{R}_{\tilde{t}}(0) = \mathbf{C}_{\tilde{t}}$, with $\mathbf{m}_{\tilde{t}}$ and $\mathbf{C}_{\tilde{t}}$ denoting the filtered mean and covariance of $\boldsymbol{\theta}_{\tilde{t}}$, respectively. This distribution is implemented in the function `exdqlmForecast()`, which is compatible with the outputs from both the MCMC and VB algorithms. Table 5 shows the input parameters used in the function, as well as their mathematical symbols seen in this section. The output of `exdqlmForecast()` includes the parameters of the forecasted distribution as well as $\mathbf{a}_{\tilde{t}}(k)$ and $\mathbf{R}_{\tilde{t}}(k)$ for the $k$ steps.

| Function | Output | Description |
|---|---|---|
| *Model construction* | | |
| `polytrendMod()` | `exdqlm` | Create $n$-th order polynomial component. |
| `seasMod()` | `exdqlm` | Create Fourier form periodic component. |
| `regMod()` | `exdqlm` | Create standard regression component. |
| `as.exdqlm()` | `exdqlm` | Coerce a `list` into an `exdqlm` object. |
| *Model fitting* – Input `exdqlm` objects | | |
| `exdqlmMCMC()` | `exdqlmMCMC` | Estimate the posterior with MCMC algorithm. |
| `exdqlmLDVB()` | `exdqlmLDVB` | Estimate the posterior with a VB approximation. |
| `exdqlmTransferLDVB()` | `exdqlmLDVB` | Estimate the transfer-function exDQLM with a VB approximation. |
| `exdqlmTransferMCMC()` | `exdqlmMCMC` | Estimate the transfer-function exDQLM with MCMC. |
| *Static fitting* – Input design matrix and response | | |
| `exalStaticLDVB()` | `exalStaticLDVB` | Estimate static exAL regression with a VB approximation. |
| `exalStaticMCMC()` | `exalStaticMCMC` | Estimate static exAL regression with MCMC. |
| *Model fit exploration* – Input `exdqlmMCMC` or `exdqlmLDVB` objects | | |
| `exdqlmForecast()` | `exdqlmForecast` | Compute $k$-step-ahead forecast quantiles. |
| `exdqlmDiagnostics()` | `exdqlmDiagnostic` | Compute & plot model diagnostics. |
| `exdqlmPlot()` | `list` | Plot posterior summary of dynamic quantile. |
| `compPlot()` | `list` | Plot posterior summary of component/parameter. |
| `exalStaticDiagnostics()` | `exalStaticDiagnostic` | Compute static fit diagnostics and comparisons. |
| *Predictive synthesis* – Input posterior predictive draw lists | | |
| `quantileSynthesis()` | `list` | Construct a unified posterior predictive distribution from anchor-specific predictive draws across fitted quantiles. |

Table 6: Overview of the core workflow functions used in this paper. Generic S3 methods defined for the fitted objects listed in the table include `plot()`, `print()`, and `summary()`.

The manuscript figures and tables are generated from scripts distributed with the article repository. These scripts rebuild the example artifacts directly from the current **exdqlm** API, and the accompanying reproducibility materials record the computing environment and manuscript outputs used in the paper.

# 4. Examples

Table 6 provides an overview of the **exdqlm** functions and the resulting objects. We demonstrate the general workflow and key features of the package through several analyses on real datasets. The first three examples illustrate dynamic workflows, while the fourth illustrates

the static exAL special case. The following examples are implemented on a personal computer with an Apple M1 processor. We begin by loading the **exdqlm** package used for all examples.

```
R> library(exdqlm)
```

## 4.1. Lake Huron

In this example, we consider the Lake Huron time series from the package **datasets**. The data are annual measurements of the level (ft) of Lake Huron from 1875 to 1972, shown in Figure 2. With this example we show how to: build a basic state-space structure, specify a prior on $\gamma$, run the MCMC algorithm, plot the estimated quantiles and forecast distributions.

To estimate the dynamic distribution of the data, we consider three quantiles, $p_0 = 0.95, 0.50,$ and 0.05. We begin by creating the state-space structure and prior used to estimate the quantiles (i.e. $\mathbf{F}_t, \mathbf{G}_t, \mathbf{m}_0$, and $\mathbf{C}_0$ discussed in Section 2). We choose to model the quantiles with a second order polynomial trend, which we construct with the `polytrendMod()` function.

```
R> model = polytrendMod(order = 2, m0 = c(mean(LakeHuron),0), C0 = 10*diag(2))
R> model

Prior mean of the state vector:
$m0
          [,1]
[1,] 579.0041
[2,]   0.0000


Prior covariance of the state vector:
$C0
     [,1] [,2]
[1,]   10    0
[2,]    0   10


Observational vector:
$FF
     [,1]
[1,]    1
[2,]    0


Evolution matrix:
$GG
     [,1] [,2]
[1,]    1    1
[2,]    0    1
```

To allow variability in the dynamic quantile, we select a single discount factor of 0.9 to define the evolution covariance matrix $\mathbf{W}_t$. Discount factors can also be optimized, an example of which we will show in Section 4.2. We place truncated Student-t priors on the skewness parameters via `PriorGamma` as discussed in Section 2.1. In the fits shown here we use `n.burn = 2000` and `n.mcmc = 3000`. The R output is shown for only the last quantile using the input `verbose = TRUE`.

```
R> M95 = exdqlmMCMC(y = LakeHuron, p0 = 0.95, model = model,
+                 df = 0.9, dim.df = 2,
+                 PriorGamma = list(m_gam = -1, s_gam = 0.1, df_gam = 1),
+                 n.burn = 2000, n.mcmc = 3000, verbose = FALSE)
R> M50 = exdqlmMCMC(y = LakeHuron, p0 = 0.50, model = model,
+                  df = 0.9, dim.df = 2,
+                  PriorGamma = list(m_gam = 0, s_gam = 0.1, df_gam = 1),
+                  n.burn = 2000, n.mcmc = 3000, verbose = FALSE)
R> M5 = exdqlmMCMC(y = LakeHuron, p0 = 0.05, model = model,
+                 df = 0.9, dim.df = 2,
+                 PriorGamma = list(m_gam = 1, s_gam = 0.1, df_gam = 1),
+                 n.burn = 2000, n.mcmc = 3000, verbose = TRUE)
```

*Representative console excerpt, abridged to the fields most relevant for the progress format:*

```
MCMC backend: C++ (mode=fast)
MCMC init: running LDVB warm start
MCMC start | burn=2000 | keep=3000 | kernel=slice | warm_start=ldvb
...
MCMC done | runtime_sec=122.179 | sigma=0.220 | gamma=5.013 | kernel=slice
```

For this time series of length 98, the free-$\sigma$ run shown above takes about 122.2 seconds to sample 5000 iterations. By default `exdqlmMCMC()` uses the package's VB routine, `exdqlmLDVB()`, to warm-start the chain, and this initialization choice can be changed independently of the subsequent MCMC kernel. For the free-$\sigma$ Lake Huron median fit, the default `slice` update was more stable than `laplace_rw` in our support comparison, so we retain `slice` here. Because the data are relatively symmetric, we do not expect the skewness parameter $\gamma$ to be far from zero at $p_0 = 0.5$; in the dedicated median run with free $\sigma$, the posterior mean of $\sigma$ is about 0.375 with a 95% credible interval $(0.298, 0.473)$, while the posterior mean of $\gamma$ is $-0.044$ with a 95% credible interval $(-0.375, 0.321)$. All retained samples in the output of `exdqlmMCMC()` are `mcmc` objects, so we can use the package **coda** to examine the trace and density plots of both $\sigma$ and $\gamma$, seen in Figure 1.

```
R> M50.trace = exdqlmMCMC(y = LakeHuron, p0 = 0.50, model = model,
+                  df = 0.9, dim.df = 2,
+                  PriorGamma = list(m_gam = 0, s_gam = 0.1, df_gam = 1),
+                  n.burn = 2000, n.mcmc = 3000, verbose = FALSE)
R> keep.idx = seq(1, length(M50.trace$samp.sigma), by = 10)
R> sigma.trace = coda::mcmc(M50.trace$samp.sigma[keep.idx], thin = 10)
R> gamma.trace = coda::mcmc(M50.trace$samp.gamma[keep.idx], thin = 10)
R> coda::traceplot(sigma.trace, main = "sigma trace")
R> coda::densplot(sigma.trace, main = "sigma density")
R> coda::traceplot(gamma.trace, main = "gamma trace")
R> coda::densplot(gamma.trace, main = "gamma density")
```

The posterior summary indicates that $\gamma$ is not distinguishable from zero, so the additional flexibility of the skewness parameter is not needed for the median fit. We therefore re-run the model with a point-mass prior on $\gamma$ at 0 using the settings `gam.init = 0` and `fix.gamma = TRUE`, which is equivalent to the setting `dqlm.ind = TRUE`. This reduces the model to the DQLM mentioned in Section 2.5.

```
R> M50 = exdqlmMCMC(y = LakeHuron, p0 = 0.50, model = model,
+                  df = 0.9, dim.df = 2,
```

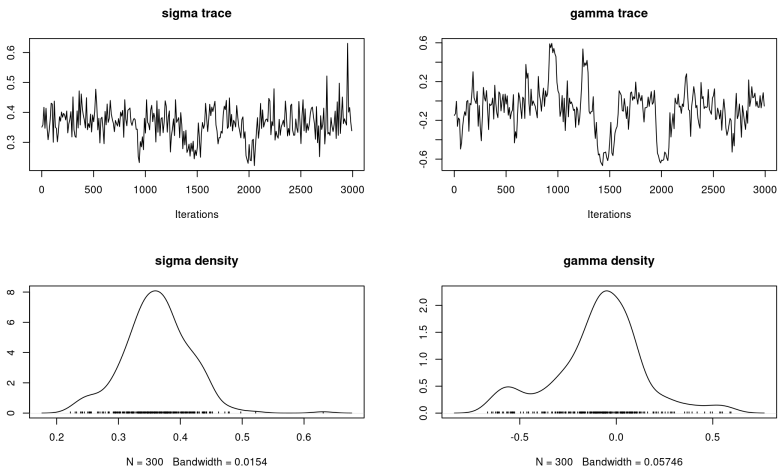


Figure 1: Example 1: Lake Huron. Trace and density plots from the dedicated free-$\sigma$ median run with 2000 burn-in iterations and 3000 retained draws. For readability, every 10th retained draw is used in the plotting step. The top row shows the posterior trace plots for $\sigma$ and $\gamma$, and the bottom row shows the corresponding posterior densities.

```
+                   gam.init = 0, fix.gamma = TRUE,
+                   n.burn = 2000, n.mcmc = 3000, verbose = TRUE)

MCMC backend: C++ (mode=fast)
MCMC init: running LDVB warm start
MCMC start | burn=2000 | keep=3000 | kernel=conjugate | warm_start=ldvb
...
MCMC done | runtime_sec=51.838 | sigma=0.430 | kernel=conjugate
```

The results are `exdqlmMCMC` objects that can be used for analysis and forecasting. First we examine the estimated quantiles with the plot method for `exdqlmMCMC` objects, seen in the top-left panel of Figure 2. This generates a plot of the data with the MAP estimates and the 95% CrIs of the dynamic quantile. Subsequent quantiles (i.e. $p_0 = 0.50, 0.05$ in this example) can be added to the plot using the function `exdqlmPlot()` with the argument `add = TRUE`.

```
R> plot(M95)
R> exdqlmPlot(M50, add = TRUE, col = "blue")
R> exdqlmPlot(M5, add = TRUE, col = "forest green")
R> legend("topright", lty = 1, col = c("purple","blue","forest green"),
+        legend = c(expression('p'[0]*'=0.95'),expression('p'[0]*'=0.50'),
+                   expression('p'[0]*'=0.05')))
```

Next, we forecast the $k = 8$ step-ahead distributions starting in 1972. To do this, we first create the observational vector (`fFF`) and evolution matrix (`fGG`) that will be used in the forecast updates. In this case both are non-time-varying and identical to those used to estimate the quantile.

```
R> fFF = model$FF
R> fGG =  model$GG
```

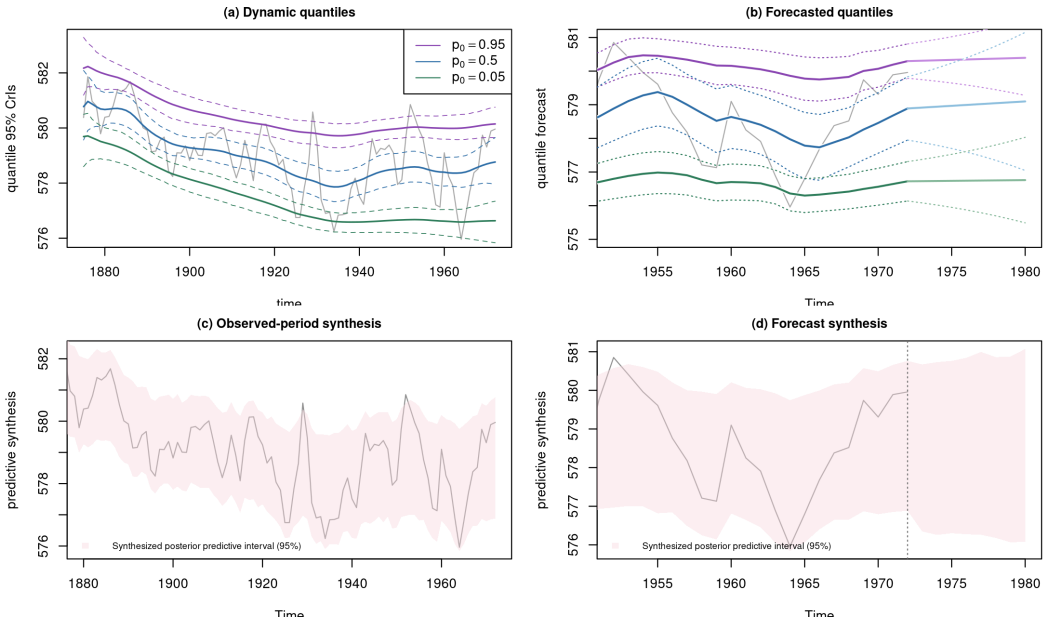


Figure 2: Example 1: Lake Huron. Top-left: MAP estimates and 95% CrIs of the estimated quantiles, plotted with the data (grey). Top-right: forecasted quantile estimates and 95% credible intervals (seen after 1972), along with the filtered quantile estimates and 95% credible intervals (seen from 1952 to 1972). Bottom-left: synthesized posterior predictive 95% interval over the observed period. Bottom-right: synthesized posterior predictive 95% interval over the forecast window.

We plot the time series data in a narrower range for closer examination and add the forecast estimates using `exdqlmForecast()` with argument `add = TRUE`. Notice we start the forecast at the last time index of the data, i.e. `start.t = length(LakeHuron)`.

```
R> plot(LakeHuron,  xlim = c(1952,1980), ylim = c(575,581),
+        col = "dark grey")
R> exdqlmForecast(start.t = length(LakeHuron), k = 8, m1 = M95,
+               fFF = fFF, fGG = fGG, plot = TRUE, add = TRUE)
R> exdqlmForecast(start.t = length(LakeHuron), k = 8, m1 = M50,
+               fFF = fFF, fGG = fGG, plot = TRUE, add = TRUE,
+               cols = c("blue","light blue"))
R> exdqlmForecast(start.t = length(LakeHuron), k = 8, m1 = M5,
+               fFF = fFF, fGG = fGG, plot = TRUE, add = TRUE,
+               cols = c("forest green","green"))
```

This generates a plot of the data with the forecasted quantile estimates and 95% credible intervals (seen after 1972), along with the filtered quantile estimates and 95% credible intervals (before 1972), for reference. Results can be seen in the top-right panel of Figure 2. The percentage in the CrIs can be modified with the parameter `cr.percent`.

When several quantile models are fitted separately, the package also allows their posterior predictive draws to be combined into a single coherent predictive distribution through `quantileSynthesis()`. Each fitted model contributes local predictive information near its target quantile, and the synthesis step interpolates across these anchor-specific draws to obtain

one posterior predictive distribution. When independently fitted quantiles induce crossing, optional monotonicity corrections based on isotonic adjustment and monotone rearrangement are available (Barlow, Bartholomew, Bremner, and Brunk 1972; Chernozhukov, Fernández-Val, and Galichon 2010). For the Lake Huron example, we apply this routine to the posterior predictive samples stored in `M5`, `M50`, and `M95` over the observed period. Since these posterior predictive draws are produced from the smoothed state draws, the lower panels of Figure 2 display the synthesized posterior predictive 95% interval over the observed and forecast windows. For the eight-step forecast horizon, we first generate future predictive draws from each fitted model using the corresponding forecasted quantile distributions together with posterior draws of $\sigma$ and $\gamma$, and then apply the same synthesis step. This provides a single predictive envelope while preserving the separate quantile-specific fits.

A direct call for the observed posterior predictive draws is given below:

```
R> syn.obs = quantileSynthesis(
+               draws_list = list(M5$samp.post.pred, M50$samp.post.pred,
+                                 M95$samp.post.pred),
+               p = c(0.05, 0.50, 0.95),
+               T_expected = length(LakeHuron))
```

The function `exdqlmForecast()` returns the forecast summaries, but not matrices of future observations. To construct those, we first sample future quantile values from the forecast distribution and then map them through `rexal()` using matched posterior draws of $\sigma$ and $\gamma$. For example, for the $p_0 = 0.05$ fit:

```
R> q.fore = sweep(matrix(rnorm(8 * n.samp), 8, n.samp), 1,
+                 sqrt(fc05$fQ), "*") + fc05$ff
R> future.M5$ydraw = vapply(1:n.samp, function(j)
+                 rexal(8, p0 = M5$p0, mu = q.fore[, j],
+                       sigma = sigma.draws[j], gamma = gamma.draws[j]),
+                 numeric(8))
```

Here `sigma.draws` and `gamma.draws` denote matched retained posterior draws from the fitted model. Analogous matrices are generated for `M50` and `M95`.

We then collect these matrices into `future.draws` and synthesize them in the same way:

```
R> future.draws = list(future.M5$ydraw, future.M50$ydraw,
+                      future.M95$ydraw)

R> syn.fore = quantileSynthesis(
+               draws_list = future.draws,
+               p = c(0.05, 0.50, 0.95),
+               T_expected = 8)
```

## 4.2. Sunspots

For our next example, we use the yearly Sunspot time series from the package **datasets**. The data are yearly counts for sunspots from 1700 to 1988, shown in Figure 3. In this example we show how to use the **dlm** package to create the state-space model, combine blocks of a state-space model, apply the VB approximation through `exdqlmLDVB()`, perform visual

model diagnostics to compare the exDQLM with the DQLM, and use those diagnostics for discount-factor selection.

The solar cycle impacts astronauts in space as well as lives and technology on earth. Of particular interest to scientists is the solar max, or period in which the sun is most active Fox (2012), therefore we will explore the 0.85 quantile in this example. Again, we begin by building the state-space structure used to estimate the quantile. Here we choose to model the quantile with a first order polynomial trend component as well as a Fourier form seasonal component that incorporates a fundamental period every 11 observations (years), as well as some of its corresponding harmonics. This results in a dynamic model with a total of 9 state parameters, 1 for the first order polynomial component and 8 for the seasonal component. The **exdqlm** functions are compatible with time-invariant `dlm` objects from the **dlm** package. For example, we create the first order trend component with the function `dlmModPoly()` from the **dlm** package, then use `as.exdqlm()` to turn the output into an `exdqlm` object.

```
R> dlm.trend.comp = dlm::dlmModPoly(1, m0 = mean(sunspot.year), C0 = 10)
R> trend.comp = as.exdqlm(dlm.trend.comp)
```

Next, we construct the seasonal component with our `seasMod` function. The duration of sunspot cycles typically follow a period of 11 years, of which we include the first 4 harmonics in the component.

```
R> seas.comp = seasMod(p = 11, h = 1:4, C0 = 10*diag(8))
```

To combine the two components into a single state-space structure, we add the two `exdqlm` objects.

```
R> model = trend.comp + seas.comp
```

The resulting evolution matrix of the combined models is printed for illustration.

```
R> model$GG
```

```
     [,1]    [,2]   [,3]    [,4]   [,5]    [,6]    [,7]    [,8]    [,9]
 [1,]    1  0.0000 0.0000  0.0000 0.0000  0.0000  0.0000  0.0000  0.0000
 [2,]    0  0.8413 0.5406  0.0000 0.0000  0.0000  0.0000  0.0000  0.0000
 [3,]    0 -0.5406 0.8413  0.0000 0.0000  0.0000  0.0000  0.0000  0.0000
 [4,]    0  0.0000 0.0000  0.4154 0.9096  0.0000  0.0000  0.0000  0.0000
 [5,]    0  0.0000 0.0000 -0.9096 0.4154  0.0000  0.0000  0.0000  0.0000
 [6,]    0  0.0000 0.0000  0.0000 0.0000 -0.1423  0.9898  0.0000  0.0000
 [7,]    0  0.0000 0.0000  0.0000 0.0000 -0.9898 -0.1423  0.0000  0.0000
 [8,]    0  0.0000 0.0000  0.0000 0.0000  0.0000  0.0000 -0.6549  0.7557
 [9,]    0  0.0000 0.0000  0.0000 0.0000  0.0000  0.0000 -0.7557 -0.6549
```

We will apply discount factors by component, i.e. a discount factor of 0.9 for the trend component of dimension 1 and a discount factor of 0.85 for the seasonal component of dimension 8. This discounting strategy can be applied with arguments `df = c(0.9,0.85)` and `dim.df = c(1,8)`.

We call the function `exdqlmLDVB()` to obtain the VB approximations. We use the routine with argument `dqlm.ind = TRUE` to estimate the DQLM, as well as with the default settings to estimate the exDQLM. In this example we set `n.samp = 3000`. The output is suppressed

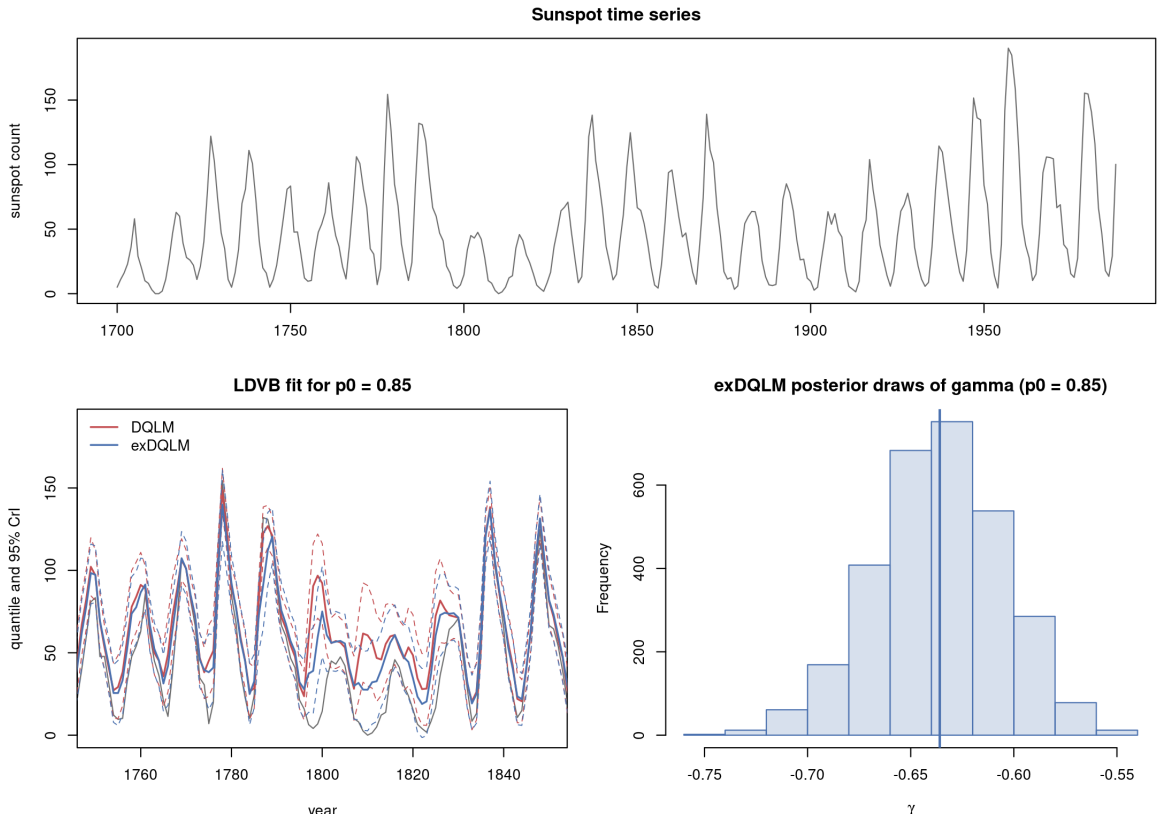


Figure 3: Example 2: Sunspots. Top: The sunspot time series from 1700 to 1988. Bottom left: MAP estimates and 95% CrIs of the estimated quantiles from the DQLM (red) and exDQLM (blue), plotted with the data (grey) from 1750 to 1850. Bottom right: Histogram of samples from the approximated posterior distribution of $\gamma$ under the exDQLM.

using input `verbose = FALSE`, however we check the run-time of each model after fitting using the `summary()` generic function. For this dataset of length 289, the DQLM VB fit completes in about 23 seconds. The corresponding exDQLM VB fit is more computationally demanding and takes about 104 seconds.

```
R> M1 = exdqlmLDVB(y = sunspot.year, p0 = 0.85, model = model,
+                  df = c(0.9,0.85), dim.df = c(1,8),
+                  dqlm.ind = TRUE, fix.sigma = FALSE,
+                  n.samp = 3000,
+                  verbose = FALSE)
R> M2 = exdqlmLDVB(y = sunspot.year, p0 = 0.85, model = model,
+                  df = c(0.9,0.85), dim.df = c(1,8),
+                  fix.sigma = FALSE,
+                  n.samp = 3000,
+                  verbose = FALSE)
R> summary(M1)

Bayesian Dynamic Quantile Regression Model (DQLM)
Number of Observations: 289
State Dimension: 9
```

```
Discount factors ( dimensions ): 0.9 ( 1 ), 0.85 ( 8 )

DQLM fitted using VB
Iterations until convergence: 67
Run-time: 22.891 seconds
```

```
R> summary(M2)
```

```
Bayesian Dynamic Quantile Regression Model (exDQLM)
Number of Observations: 289
State Dimension: 9
Discount factors ( dimensions ): 0.9 ( 1 ), 0.85 ( 8 )

exDQLM fitted using LDVB
Parameters estimated with LD: gamma sigma
Iterations until convergence: 200
Run-time: 104.092 seconds
```

We can visualize the differences between the two resulting dynamic quantiles using the function `exdqlmPlot()`, shown in the lower-left panel of Figure 3. For clarity we limit that panel to years 1750 to 1850.

```
R> plot(sunspot.year, xlim = c(1750,1850), col = "dark grey",
+       ylab = "quantile 95% CrIs")
R> exdqlmPlot(M1, add = TRUE, col = "red")
R> exdqlmPlot(M2, add = TRUE, col = "blue")
```

To examine whether the added flexibility of the exDQLM is significant, we can also visualize the approximate posterior distribution of the skewness parameter $\gamma$ by plotting the samples from the corresponding variational distribution, also seen in Figure 3.

```
R> hist(M2$samp.gamma,xlab=expression(gamma),main="")
```

The approximate distribution is clearly distinct from zero, thus we expect a better model fit from the exDQLM.

To further examine the two models, we use the function `exdqlmDiagnostics()` to plot the model diagnostics discussed in Section 3. The results are seen in Figure 4. The QQ-plot and the ACF of the standard forecast errors both favor the exDQLM.

```
R> exdqlmDiagnostics(M1, M2, cols = c("red","blue"))
```

Finally, the function `exdqlmDiagnostics()` can also be used for model selection. Say we want to check whether the discount factor we used for the seasonal component is reasonable under a predictive scoring rule. We can create a list of possible discount factors, quickly run the VB approximation for each possible model, and compare the resulting CRPS and KL values. In practice we recommend selecting the discount factors with the CRPS and using the KL divergence as a complementary calibration check. A simple example is seen below.

```
R> possible.dfs = cbind(0.9,seq(0.85,1,0.05))
R> possible.dfs
```

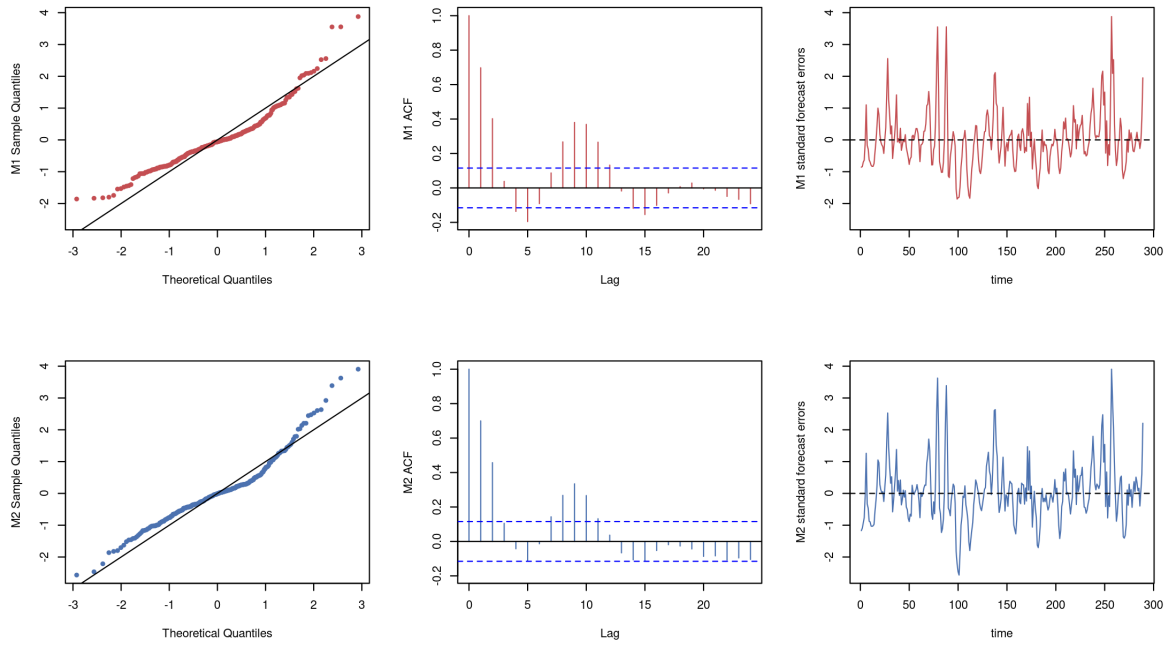


Figure 4: Example 2: Sunspots. The QQ-plots (left column), ACF plots (center column) and standard forecast errors (right column) from the DQLM (red) and exDQLM (blue).

```
     [,1] [,2]
[1,]  0.9 0.85
[2,]  0.9 0.90
[3,]  0.9 0.95
[4,]  0.9 1.00
```

```
R> metrics <- matrix(NA_real_, nrow(possible.dfs), 2,
+                    dimnames = list(NULL, c("CRPS","KL")))
R> ref.samp = rnorm(length(sunspot.year))
R> for(i in 1:nrow(possible.dfs)){
+   temp.M2 = exdqlmLDVB(y = sunspot.year, p0 = 0.85, model = model,
+                  df = possible.dfs[i,], dim.df = c(1,8),
+                  sig.init = 2, fix.sigma = FALSE,
+                  n.samp = 3000, verbose = FALSE)
+   temp.check = exdqlmDiagnostics(temp.M2, plot = FALSE,
+                              ref = ref.samp)
+   metrics[i,] = c(temp.check$m1.CRPS, temp.check$m1.KL)
+ }
R> # optimal dfs based off CRPS
R> possible.dfs[which.min(metrics[, "CRPS"]),]
```

```
[1] 0.90 0.85
```

On this coarse grid, the CRPS and KL criteria agree and both select the seasonal discount factor 0.85, so the choice used above is retained.

The runtime comparisons below were obtained under the default implementation used by the

| model | method | runtime (s) | KL | CRPS | pplc |
|---|---|---|---|---|---|
| DQLM | VB | **22.89** | **0.144** | 8.624 | 3906.5 |
| DQLM | MCMC | 273.68 | 0.153 | **7.708** | **3051.2** |
| exDQLM | VB | **104.09** | **0.078** | 6.416 | 2350.3 |
| exDQLM | MCMC | 343.14 | 0.126 | **5.075** | **2189.8** |

Table 7: Example 2: representative dynamic benchmark for the DQLM and exDQLM fits at $p_0 = 0.85$. Each row reports runtime together with the KL divergence, the mean CRPS from posterior predictive draws, and the posterior predictive loss criterion. The VB rows use `n.samp = 3000`. The MCMC rows use 2000 burn-in iterations and 3000 retained draws. Within each model block, bold entries indicate the smaller value for that metric.

dynamic routines. Exact hardware, software, seed, and dataset details are recorded in the accompanying reproducibility materials.

Table 7 gives a compact dynamic runtime-and-quality comparison under matched 3000-draw summaries. For the simpler DQLM, the VB fit remains much faster than MCMC and attains the smaller KL divergence, whereas the MCMC fit achieves smaller values for the CRPS and the pplc. For the more flexible exDQLM, the VB fit is also faster than the corresponding MCMC run, but the comparison is again mixed: the VB fit attains the smallest KL divergence in the table, while the MCMC fit is preferred by both the CRPS and the pplc. Within each inferential method, the exDQLM improves the reported diagnostics relative to the DQLM. These results again reinforce the importance of reporting runtime together with predictive criteria rather than runtime alone: the practical tradeoff depends on both the model and the inferential method being compared.

### 4.3. Big Tree water flows

For our final example, we consider the average monthly natural water flow (cubic feet per second) at the Big Tree gauge of the San Lorenzo river in Santa Cruz County, CA, obtained by averaging daily USGS observations within each calendar month (U.S. Geological Survey 2016). In the article workflow we construct this monthly series directly from the staged daily USGS file rather than relying on a packaged monthly flow dataset. To keep the covariate structure of the original example, we pair this monthly USGS-derived series with the monthly Niño 3.4 index (Rayner, Parker, Horton, Folland, Alexander, Rowell, Kent, and Kaplan 2003) supplied by the package. Because the package `nino34` series ends in April 2021, we restrict attention to the overlapping window from January 1987 through April 2021, for a total of 412 observations. With this example we show how to build a state-space structure manually for a dynamic quantile regression, apply our transfer-function exDQLM, examine individual components of the estimated quantiles, and perform quantitative model diagnostics to compare models.

```
R> bt.daily = read.csv("big_trees_daily_usgs_ppt_soil.csv")
R> bt.daily$date = as.Date(bt.daily$date)
R> bt.daily = bt.daily[order(bt.daily$date), c("date","usgs_cfs")]
R> month.id = format(bt.daily$date, "%Y-%m")
R> flow.monthly.df = aggregate(bt.daily["usgs_cfs"],
+                              by = list(month = month.id), FUN = mean)
R> flow.monthly = ts(flow.monthly.df$usgs_cfs, start = c(1980,1),
+                    frequency = 12)
```

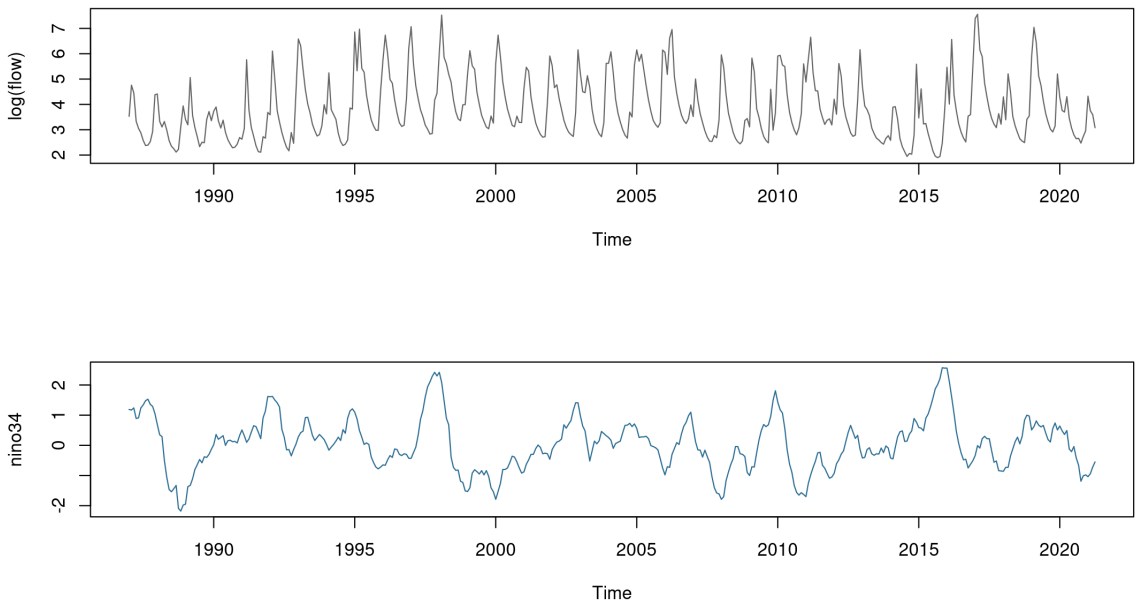


Figure 5: Example 3: Big Tree water flow. Top: average monthly water flow at the Big Tree gauge of the San Lorenzo river in Santa Cruz County, CA, plotted on the log scale after aggregating the daily series to monthly means. Bottom: monthly values of Niño 3.4. Both series span January 1987 through April 2021.

```
R> flow.monthly = window(flow.monthly, end = c(2021,4))
R> nino.monthly = window(nino34, start = start(flow.monthly),
+                        end = end(flow.monthly))
R> flow.fit = flow.monthly
R> nino.input = nino.monthly
```

For this particular dataset, the interest is in low quantiles as a possible indicator of drought, thus we consider the $p_0 = 0.15$ quantile. To investigate climatological influence on this low quantile, we include the monthly Niño 3.4 index as an input covariate in both models. The dynamic regression specification uses Niño 3.4 directly as a time-varying regressor, while the transfer-function model uses the same contemporaneous monthly input and lets the transfer-function structure capture persistence through $\zeta_t$ and $\lambda$. The fitted comparison below therefore uses the full overlapping sample from January 1987 through April 2021, for a total of 412 observations. To avoid modeling problems caused by the scale and extremes in the Big Tree dataset, we fit the following models to the log monthly flow series shown in Figure 5.

In addition to the component that captures the relationship with the Niño 3.4 input, we include a first-order polynomial trend component together with a Fourier-form seasonal component with period 12. To mirror the annualized seasonal structure used in the daily redesign, we include harmonics $h = 1$, $h = 2$, and $h = 0.1469118636$, corresponding to annual, semiannual, and approximately 6.8-year cycles on the monthly scale.

```
R> trend.comp = polytrendMod(1, m0 = 3, C0 = 0.1)
R> seas.comp = seasMod(p = 12, h = c(1, 2, 0.1469118636),
```

```
+                     C0 = diag(1,6))
R> model = trend.comp + seas.comp
```

Next, we create the dynamic regression component for our first model using the monthly Niño 3.4 series directly. Combining this with the trend and seasonal components gives our complete state-space structure for the dynamic quantile regression model.

```
R> reg.comp = regMod(nino.input, m0 = 0, C0 = matrix(1,1,1))
R> model.w.reg = model + reg.comp
```

To allow all components to evolve gradually over time while keeping the fitted paths smooth, we set every discount factor in both models to 0.97. In particular, we use 0.97 for the trend block, the seasonal block, the monthly Niño 3.4 regression block, and the transfer-function block.

The transfer-function model follows the structure detailed in Section 2.4. For this illustration we use `exdqlmTransferLDVB()` to obtain a fast approximate fit on the monthly series, although the corresponding exact posterior workflow is also available through `exdqlmTransferMCMC()`. On the coarse candidate grid used here, we use the KL divergence as a calibration-oriented screening criterion to select the transfer-function decay rate, and then assess the resulting fitted models with the broader diagnostic comparison reported below. This screen selects a decay rate of 0.50.

```
R> possible.lams = c(0.1,0.3,0.5,0.7,0.85)
R> KLs <- vector("numeric")
R> ref.samp = rnorm(length(flow.fit))
R> for(i in 1:length(possible.lams)){
R>   temp.M2 = exdqlmTransferLDVB(y = log(flow.fit), p0 = 0.15, model = model,
+                               df = c(0.97,0.97), dim.df = c(1,6),
+                               X = nino.input, tf.df = c(0.97),
+                               lam = possible.lams[i],
+                               tf.m0 = c(0,0),
+                               tf.C0 = diag(c(0.1,0.005),2),
+                               sig.init = 0.1, gam.init = - 0.1,
+                               fix.sigma = FALSE,
+                               tol = 0.05, verbose = FALSE)
R>   temp.check = exdqlmDiagnostics(temp.M2, plot = FALSE,
+                             ref = ref.samp)
R>   KLs = c(KLs,temp.check$m1.KL)
R> }
R> # optimal lambda based off KL divergence
R> possible.lams[which.min(KLs)]
```

```
[1] 0.5
```

With these discount factors and optimal $\lambda$, we estimate the dynamic quantile regression model (`M1`) and transfer function model (`M2`).

```
R> M1 = exdqlmLDVB(y = log(flow.fit), p0 = 0.15, model = model.w.reg,
+                df = c(0.97,0.97,0.97), dim.df = c(1,6,1),
+                sig.init = 0.1, gam.init = - 0.1,
+                fix.sigma = FALSE,
+                tol = 0.05, verbose = FALSE)
```

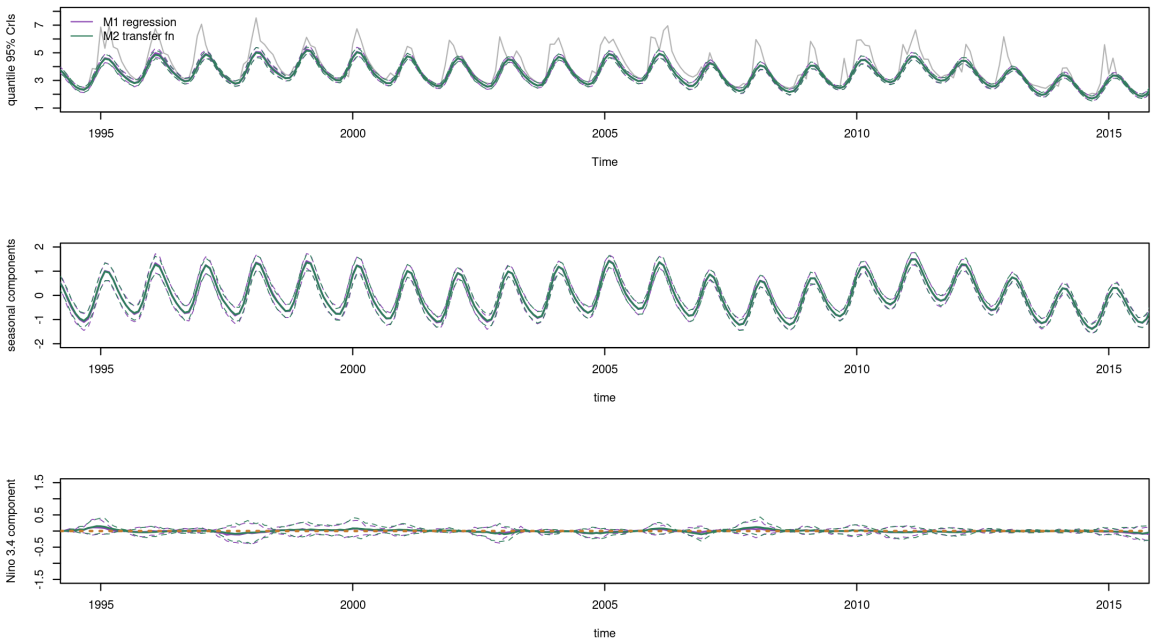


Figure 6: Example 3: Big Tree water flow. Top: MAP estimates and 95% CrIs of the estimated quantiles from the models with the dynamic regression component (purple) and transfer-function component (green), plotted with the data (grey) from 1995 to 2015. Middle: MAP estimates and 95% CrIs of the combined seasonal components implied by the annual, semiannual, and low-frequency harmonics. Bottom: MAP estimates and 95% CrIs of the components that model the quantile association with Niño 3.4. For $M_1$, the purple curve is the regression contribution of the monthly Niño 3.4 covariate. For $M_2$, the green curve is the transfer-function contribution. The horizontal orange dotted line is at zero for reference.

```
R> M2 = exdqlmTransferLDVB(y = log(flow.fit), p0 = 0.15, model = model,
+                          df = c(0.97,0.97), dim.df = c(1,6),
+                          X = nino.input, tf.df = c(0.97), lam = 0.50,
+                          tf.m0 = c(0,0),
+                          tf.C0 = diag(c(0.1,0.005),2),
+                          sig.init = 0.1, gam.init = - 0.1,
+                          fix.sigma = FALSE,
+                          tol = 0.05, verbose = FALSE)
```

To visualize the differences between the two resulting dynamic quantiles we use the function `exdqlmPlot`, the results of which are found in Figure 6. For clarity we limit the figure to the period 1995 to 2015.

```
R> plot(log(flow.fit), col = "grey", ylim = c(1,8), xlim=c(1995,2015),
+       ylab = "quantile 95% CrIs")
R> exdqlmPlot(M1, add = TRUE)
R> exdqlmPlot(M2, add = TRUE, col = "forest green")
```

The resulting quantiles exhibit visible differences. To identify from where these differences arise, we can use the function `compPlot` to examine the model components more closely. We

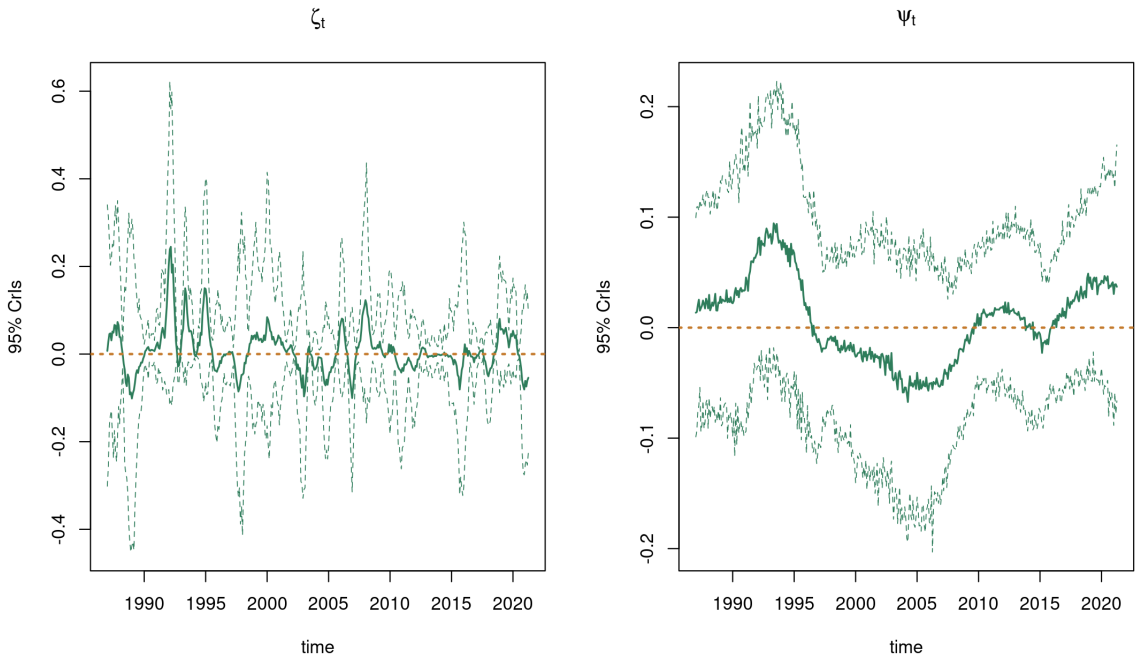


Figure 7: Example 3: Big Tree water flow. MAP estimates and 95% CrIs of the transfer-function states $\zeta_t$ and $\psi_t$. Horizontal dotted orange lines are at zero for reference.

examine the combined seasonal block first. Because the seasonal structure now contains three harmonics, the seasonal contribution is formed by the second through seventh elements of the state vector. The result is seen in Figure 6, again limited to the years 1995 to 2015.

```
R> plot(NA, ylim = c(-2,2), xlim = c(1995,2015),
+         ylab = "seasonal components")
R> compPlot(M1, index = 2:7, add = TRUE)
R> compPlot(M2, index = 2:7, add = TRUE, col = "forest green")
```

Although the seasonal components of the two models exhibit differences in amplitude, more substantial differences appear in the Niño 3.4-related components. Again, we use the function `compPlot` to visualize these components seen in Figure 6 from 1995 to 2015. For the regression model, the plotted contribution is simply the dynamic regression contribution of the monthly Niño 3.4 covariate.

```
R> plot(NA, ylim = c(-2,2), xlim = c(1995,2015),
+          ylab = "Nino 3.4 components")
R> compPlot(M1, index = 8, add = TRUE)
R> compPlot(M2, index = 8:9, add = TRUE, col = "forest green")
R> abline(h = 0, col = "orange", lty = 3, lwd = 2)
```

For the transfer-function model, `M2`, it is also of interest to look at the estimates of parameters $\zeta_t$ and $\psi_t$ directly. Because these parameters augment the state-space model as discussed in Section 2.4, the function `compPlot` can also be used to look at them directly via the parameter `just.theta = TRUE`. This setting returns the dynamic posterior estimates of a single index

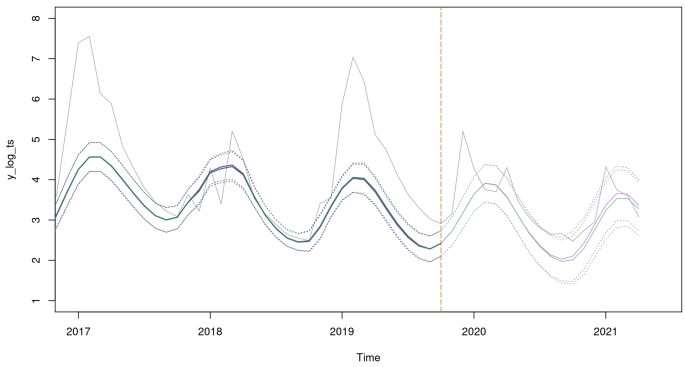


Figure 8: Example 3: Big Tree water flow. 18-step-ahead quantile forecast beginning October 2019 and extending through April 2021, overlaid on the log Big Tree water flow time series. The vertical dot-dashed line marks the start of the forecast period. Solid lines indicate mean estimates and dotted lines 95% CrI, with the filtered estimates shown before the forecast origin and the forecast estimates beyond it. Estimates from $M_1$, the quantile regression model, are seen in purple/pink. Estimates from $M_2$, the transfer function model, are seen in green.

of the state vector $\boldsymbol{\theta}_t$ (or $\tilde{\boldsymbol{\theta}}_t$ in the case of the transfer-function model). Here the indices 8 and 9 correspond to $\zeta_t$ and $\psi_t$, respectively. Results can be found in Figure 7.

```
R> compPlot(M2, index = 8, col= "forest green",
+             add = FALSE, just.theta = TRUE)
R> abline(h = 0, col = "orange", lty = 3, lwd = 2)
R> title(expression(zeta[t]))
R> compPlot(M2, index = 9, col = "forest green",
+             add = FALSE, just.theta = TRUE)
R> abline(h = 0, col = "orange", lty = 3, lwd = 2)
R> title(expression(psi[t]))
```

To further understand these transfer-function parameters, we examine the median number of time steps until the effect of Niño 3.4 on the 0.15 quantile is less than or equal to 1e−3, as discussed in Section 2.4. Here the effect is $|\psi_{t-1}X_t|$, where $X_t$ denotes the contemporaneous monthly Niño 3.4 input. A median value of about 3.76 suggests that the transfer effect lasts on the order of four months.

```
R> M2$median.kt
```

```
[1] 3.758485
```

Because the Niño 3.4 series is available through April 2021, we use the final 18 months of the overlapping sample to illustrate the forecasting workflow; the results are shown in Figure 8.

```
R> exdqlmForecast(start.t = length(flow.monthly)-18, k = 18, m1 = M1)
R> exdqlmForecast(start.t = length(flow.monthly)-18, k = 18, m1 = M2,
+                   add = TRUE, cols = c("forest green","green"))
```

| Criterion | M1 regression | M2 transfer function |
|---|---|---|
| CRPS | **0.349** | 0.357 |
| pplc | 130.81 | **123.37** |
| KL | **0.0565** | 0.0570 |

Table 8: Example 3: Big Tree water flow. Diagnostic comparison between the dynamic regression model (M1) and the transfer-function model (M2). Lower values are preferred for all three criteria. Here CRPS assesses the full predictive distribution, pplc assesses target-quantile posterior predictive loss, and KL summarizes one-step-ahead calibration discrepancy. Bold entries indicate the smaller value in each row.

Finally, we use the function `exdqlmDiagnostics()` to compute the KL divergence, CRPS, and pplc discussed in Section 3. Table 8 summarizes the resulting diagnostic comparison.

```
R> checks.out = exdqlmDiagnostics(M1, M2,
+                                 cols = c("purple","orange"), plot = FALSE)
```

With this simpler monthly Niño 3.4 specification, the resulting diagnostics are again not unanimous. The direct regression model attains the smaller KL divergence and CRPS, whereas the transfer-function model achieves the smaller pplc. The differences are fairly modest, so we view the two specifications as offering closely competing summaries of the same low-quantile dynamics: the direct regression is slightly favored by the calibration-oriented and full-distribution criteria, while the transfer-function model is favored by the target-quantile loss criterion.

### 4.4. Static exAL regression on a simulated heteroskedastic example

To illustrate the static routines under sparse shrinkage, we consider a correlated Gaussian regression benchmark for which the target conditional quantile is known exactly. Let

$$\mathbf{x}_i \sim N_8(\mathbf{0}, \Sigma_x), \qquad (\Sigma_x)_{jk} = 0.5^{|j-k|},$$

with sparse signal

$$\boldsymbol{\beta}^\star = (3,\ 1.5,\ 0,\ 0,\ 2,\ 0,\ 0,\ 0)^\top.$$

We use the following benchmark data-generating mechanism. For each target quantile level $p_0$, we generate

$$Y_i^{(p_0)} = \mathbf{x}_i^\top \boldsymbol{\beta}^\star + \sigma_\varepsilon\{Z_i - \Phi^{-1}(p_0)\}, \qquad Z_i \sim N(0,1), \qquad \sigma_\varepsilon = 1.5.$$

Then

$$Q_Y(p_0 \mid \mathbf{x}_i) = \mathbf{x}_i^\top \boldsymbol{\beta}^\star,$$

so the true $p_0$-quantile is the same sparse linear signal at each fitted quantile level. We use $n = 160$ training observations and an independent holdout sample of size 800, and fit separate static exAL models at $p_0 \in \{0.05, 0.25, 0.50\}$. Both fits use the regularized horseshoe (RHS) prior of Nishimura and Suchard (2023), implemented in the package through `beta_prior = "rhs_ns"`, with $\tau_0 = 0.15$, fixed slab $\zeta^2 = 9$, and an unshrunk intercept. We fit a VB approximation with `exalStaticLDVB()` using `n.samp = 3000`, and an MCMC

fit with `exalStaticMCMC()` using `n.burn = 2000` and `n.mcmc = 3000`; the MCMC routine uses its default `slice` kernel and is warm-started from the VB fit. The lower-tail VB fit at $p_0 = 0.05$ converges more slowly, so we allow a larger VB iteration budget there. For reproducibility, we fix the dataset seed at `20260714`. Setting `dqlm.ind = TRUE` would recover the AL special case discussed in Section 2, but here we focus on the general static exAL model under sparse RHS shrinkage.

```
R> Sigma.x = 0.5 ^ as.matrix(dist(1:8))
R> beta.true = c(3, 1.5, 0, 0, 2, 0, 0, 0)
R> rhs.ctrl = list(tau0 = 0.15, a_zeta = 2, b_zeta = 9,
                   zeta2_fixed = 9, shrink_intercept = FALSE)
R> set.seed(20260714)
R> sim.y = function(X.raw, p0) {
      drop(X.raw %*% beta.true) + 1.5 * (rnorm(nrow(X.raw)) - qnorm(p0))
    }
R> X.raw = MASS::mvrnorm(160, mu = rep(0, 8), Sigma = Sigma.x)
R> X = cbind(1, X.raw)
R> p.grid = c(0.05, 0.25, 0.50)
R> y.train = lapply(p.grid, function(p0) sim.y(X.raw, p0))
R> M.ldvb = lapply(p.grid, function(p0) exalStaticLDVB(
+                    y = y.train[[which(p.grid == p0)]], X = X, p0 = p0,
+                    beta_prior = "rhs_ns",
+                    beta_prior_controls = rhs.ctrl,
+                    max_iter = ifelse(p0 == 0.05, 420, 260),
+                    n.samp = 3000,
+                    tol = 1e-4, verbose = FALSE))
R> M.mcmc = lapply(p.grid, function(p0) exalStaticMCMC(
+                    y = y.train[[which(p.grid == p0)]], X = X, p0 = p0,
+                    beta_prior = "rhs_ns",
+                    beta_prior_controls = rhs.ctrl,
+                    n.burn = 2000, n.mcmc = 3000,
+                    init.from.vb = TRUE,
+                    verbose = FALSE))
```

For static fits, the helper function `exalStaticDiagnostics()` summarizes fitted quantiles on a common design matrix and, when a reference quantile is supplied, reports truth-recovery metrics directly. For example, on an independent holdout design:

```
R> X.hold.raw = MASS::mvrnorm(800, mu = rep(0, 8), Sigma = Sigma.x)
R> X.hold = cbind(1, X.hold.raw)
R> ref.hold = drop(X.hold %*% c(0, beta.true))
R> y.hold = sim.y(X.hold.raw, 0.25)
R> exalStaticDiagnostics(M.ldvb[[2]], M.mcmc[[2]],
+                   X = X.hold, y = y.hold, ref = ref.hold,
+                   plot = FALSE)
```

Figure 9 shows that both methods recover the active coefficients and shrink the null coefficients toward zero across the lower tail, lower-middle quantile, and median. The lower-tail fit at $p_0 = 0.05$ is the most demanding case, yet both methods still recover the sparse signal cleanly. Across all three quantile levels, the visual differences between VB and MCMC are modest, with the most noticeable separation appearing in the lower tail.

Table 9 reinforces the visual comparison. The two methods remain competitive across all three quantile levels, but the relative advantage depends on the metric. At $p_0 = 0.05$, the VB fit is

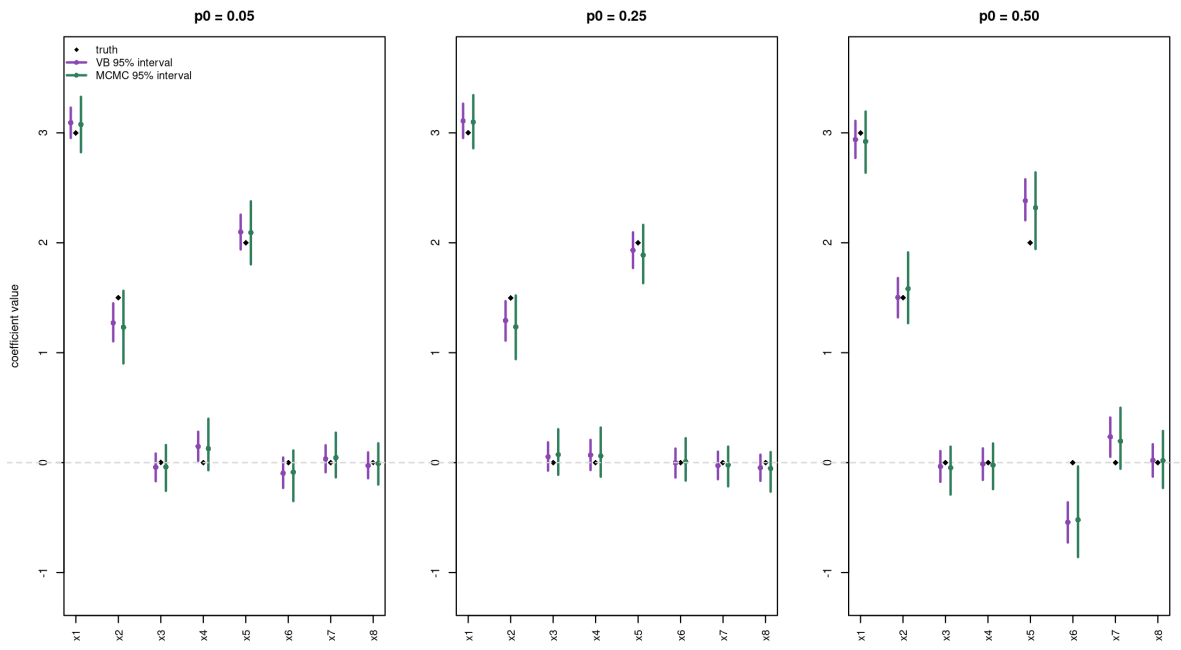


Figure 9: Example 4: sparse static exAL regression under the correlated Gaussian benchmark with the RHS prior. Black diamonds denote the true non-intercept coefficients, orange intervals denote VB 95% posterior intervals, and blue intervals denote MCMC 95% posterior intervals. Panels correspond to p0 = 0.05, 0.25, and 0.50.

| $p_0$ | method | runtime (s) | active RMSE | null MAE | holdout quantile RMSE |
|---|---|---|---|---|---|
| 0.05 | VB | **21.52** | **0.153** | 0.069 | 0.471 |
| | MCMC | 94.20 | 0.170 | **0.061** | **0.413** |
| 0.25 | VB | **10.92** | **0.141** | **0.040** | 0.478 |
| | MCMC | 96.51 | 0.176 | 0.043 | **0.476** |
| 0.50 | VB | **5.52** | 0.224 | 0.169 | 0.476 |
| | MCMC | 81.81 | **0.195** | **0.161** | **0.439** |

Table 9: Example 4: runtime and sparse-signal recovery metrics for the static exAL fits under the RHS prior. The VB rows use `n.samp = 3000`, and the MCMC rows use 2000 burn-in iterations and 3000 retained draws. The active RMSE is computed over the three nonzero coefficients, the null MAE is computed over the five zero coefficients, and the holdout quantile RMSE is computed against the true target quantile on an independent holdout sample of size 800. Within each quantile block, bold entries indicate the smaller value for that metric; ties are bolded in both rows.

slightly better on the active coefficients, whereas MCMC is better on the null coefficients and the holdout quantile RMSE. At $p_0 = 0.25$, the VB fit is slightly better on both the active-signal and null-coefficient metrics, while MCMC has a marginally smaller holdout quantile RMSE. At $p_0 = 0.50$, MCMC again holds a small advantage on the active-signal, null-coefficient, and holdout metrics. The VB fit is nevertheless much faster in all three cases, with runtime

reductions of roughly 4-fold at $p_0 = 0.05$, 9-fold at $p_0 = 0.25$, and 15-fold at $p_0 = 0.50$. In this benchmark, the VB approximation offers meaningful computational savings as a first-pass analysis while still recovering the sparse signal well, but this speed comes with the usual variational tradeoff that posterior uncertainty quantification may be less faithful than the exact MCMC baseline. Accordingly, the static RHS specification supports both roles cleanly: VB for rapid screening and model exploration, and static MCMC as the reference method when higher-fidelity posterior uncertainty quantification is required.

## 5. Conclusion

In this paper we presented the R package **exdqlm** for Bayesian estimation and analysis of flexible dynamic quantile linear models. The package provides exact posterior inference via MCMC and fast approximate posterior inference via VB for dynamic state-space quantile models, while also supporting transfer-function exDQLMs, static exAL quantile regression with shrinkage priors, forecasting, model diagnostics, and post hoc posterior-predictive synthesis across separately fitted quantiles. Together, these functions provide a practical framework for dynamic and static quantile modeling beyond the AL special case while preserving a computationally convenient latent-variable representation.

The examples illustrate the main workflows supported by the package. The dynamic examples show how **exdqlm** can be used to construct state-space models, estimate dynamic quantiles, compare the exDQLM with its AL special case, examine transfer function effects, and evaluate model fit through diagnostics and forecasting. The static example shows how the same framework can be used for sparse quantile regression with both VB and MCMC, with VB serving as a fast first-pass approximation and MCMC providing the reference analysis when more faithful posterior uncertainty quantification is required. Viewed together, these examples highlight the two software contributions that most clearly distinguish the package within the current ecosystem: Bayesian dynamic state-space quantile modeling and a unified workflow that extends from static regularized exAL regression to post hoc predictive synthesis.

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